

FEMS

Forest Exploitation Management System

USER GUIDE

How to use the application: your account and role, every screen, and the workflows that carry a forest file from application to inspection, payment and compliance.

[Mobile app](#) · [REST API](#) · [Offline field capture](#) · [AI alerts](#)

This guide covers the FEMS mobile app — Android, iOS and the web — and the REST API behind it: your account and role, every screen, and the workflows that carry a forest file from application to inspection, payment and compliance.

Generated from docs/USER-GUIDE.md · 03 October 2026

Contents

1	What FEMS does	4
2	Getting started	4
2.1	Running FEMS (for the person who installs it)	4
2.2	Signing in	5
2.3	Demonstration accounts	5
2.4	If you cannot sign in	6
3	Accounts, roles and permissions	6
3.1	The seven roles	6
3.2	What each role sees	6
3.3	Permissions	7
4	The mobile app, screen by screen	7
4.1	Home	7
4.2	Map	7
4.3	Permits	8
4.4	Payments	8
4.5	Field	8
4.6	Alerts	9
4.7	Assistant	9
4.8	Sync (field operators)	9
4.9	Profile	9
4.10	Public explore	9
5	Step-by-step workflows	9
5.1	Apply for an exploitation permit — company representative	9
5.2	Review an application — government forest officer	10
5.3	Pay the fee and get a receipt — company representative	10
5.4	Declare a harvesting activity — field operator	10
5.5	Record a field observation — any field role	11
5.6	Run an inspection — forest inspector	11
5.7	Open and resolve a violation — environmental officer	11

5.8 Handle an AI alert — officer or administrator	12
5.9 Generate a report	12
6 Working in the field: offline, GPS and photos	12
6.1 Offline capture	12
6.2 GPS and evidence quality	13
7 Artificial intelligence in FEMS	13
7.1 Alerts	13
7.2 The Forest Assistant	13
7.3 Analysis	14
8 Reports and documents	14
9 Administration	14
9.1 User accounts	14
9.2 Roles and permissions	14
9.3 System settings	14
9.4 System monitoring	15
9.5 The audit trail	15
10 Using the API directly	15
11 Troubleshooting	17
12 Glossary	18
Permit lifecycle	18
Violation lifecycle	19
Inspection lifecycle	19
Activity lifecycle	19

Forest Exploitation Management System.

This guide is written for the people who use FEMS every day — forestry officers, field inspectors and operators, company representatives, environmental officers and administrators — not for the people who build it. For installation and architecture, see the **README** (`../README.md`); for the API, see **Postman collection & screenshots** (`./postman/README.md`).

1 What FEMS does

FEMS is the operational record of a forest estate. It answers three questions:

- **What is there?** — forests, management units (zones), species, protected areas, inventory plots, and where all of it sits on a map.
- **What is happening?** — who is allowed to harvest what, where, until when and in what volume; what actually got cut, moved and replanted; who inspected it and what they found.
- **What went wrong, and what is owed?** — violation cases, penalties, remediation, and whether the fees and royalties were paid.

It is made of three parts:

Part	What it is	Who touches it
API (<code>fems-backend</code> , NestJS + Prisma + MySQL)	The single source of truth. Every rule is enforced here.	Developers, integrators, anyone running Postman
Mobile app (<code>fems-mobile</code> , Expo / React Native)	The tool people carry. Runs on Android, iOS and in a browser.	Everyone
Database (MySQL)	The records.	Administrators

Everything the app can do, the API can do — the app is only a client. That matters for two reasons: you can always fall back to the API (or Postman) for a bulk action, and no permission is ever enforced only in the app.

2 Getting started

2.1 Running FEMS (for the person who installs it)

```
git clone <your fork> FEMS && cd FEMS

npm run setup                # install backend + mobile dependencies

cd fems-backend
cp .env.example .env         # then edit DATABASE_URL to point at your MySQL
npm run prisma:generate
npm run db:migrate           # create the schema
npm run db:seed              # roles, permissions and demonstration data
npm run start:dev            # API on http://localhost:3000/api/v1
```

The mobile app:

```
npm --prefix fems-mobile run start    # Expo — scan the QR code with Expo Go
npm --prefix fems-mobile run web      # or open it in a browser
```

The app finds the API automatically: on the web it uses the page origin, on a phone it uses the address that served the JavaScript bundle. To point it somewhere else, set `EXPO_PUBLIC_API_URL` before starting Expo:

```
EXPO_PUBLIC_API_URL=https://fems.example.org/api/v1 npm --prefix fems-mobile run start
```

Interactive API documentation (Swagger) is served at **`http://localhost:3000/api/v1/docs`**. It is the fastest way to see what a route expects.

2.2 Signing in

Open the app and sign in with your work email and password. The first screen after sign-in is your **Home** dashboard; the tabs along the bottom change with your role (see §3).

If you have no account yet, **Create account** on the welcome screen registers you. Self-registration offers two roles only — Forest explorer (public data) and Company representative (your company's file). Officer, inspector, operator and administrator accounts are created by the administration, because they carry permissions over other people's records.

If your account has never been verified, you will be asked for the code sent to your email before you can do anything else. Ask an administrator to resend it if it expired.

2.3 Demonstration accounts

A seeded database ships with one account per role. **Password for all of them:** `FemsDemo#2026`

Email	Role	What it is for
<code>demo.admin@fems.cm</code>	Administrator	Everything, including users, roles and system settings
<code>demo.officer@fems.cm</code>	Government forest officer	The permit register: review, approve, activate
<code>demo.environment@fems.cm</code>	Environmental officer	Inspections, observations, violation cases
<code>demo.inspector@fems.cm</code>	Forest inspector	Field inspections and site verification
<code>demo.operator@fems.cm</code>	Field operator	Activities, observations, offline capture
<code>demo.company@fems.cm</code>	Company representative	Its own permits, activities and payments
<code>demo.explorer@fems.cm</code>	Forest explorer	Public forest data, the map and the Assistant

There are also four additional company accounts sharing one operator, useful for seeing how scope works: `demo.cooperative@...`, `demo.efc@...`, `demo.bsc@...`, `demo.cwpi@...`.

Demonstration records are flagged **DEMO** in the interface and `isDemo: true` in the API. Never seed a production database with them.

2.4 If you cannot sign in

Symptom	Cause	What to do
Invalid credentials, and you are sure of the password	Five failed attempts lock the account for 15 minutes (the counter resets on a successful sign-in)	Wait 15 minutes, or ask an administrator to set your status back to Active
The app says it cannot reach the server	The API is not running, or the app is pointed at the wrong address	Check <code>http://<host>:3000/api/v1/system/health</code> in a browser, then review <code>EXPO_PUBLIC_API_URL</code>
Email not verified	The account was created but never activated	Use Resend the code on the verification screen
<code>429 Too many requests</code>	More than 10 sign-ins per minute from the same address	Wait a minute and retry
The app opens but every list is empty	Signed in with a role that has no data in scope	Check your role on Profile → Roles and scope

3 Accounts, roles and permissions

3.1 The seven roles

FEMS ships with seven roles. A person can hold more than one; the navigation always follows the **highest** role they hold.

Role	Works on	Typical day
Administrator	Everything	Creates accounts, grants roles, configures the system, reads the audit trail
Government forest officer	The permit register	Reviews applications, requests revisions, approves, suspends, revokes
Environmental officer	Compliance and environment	Plans inspections, records observations, opens and resolves violation cases, follows remediation
Forest inspector	Field verification	Runs inspections against a checklist, records GPS and photos, submits findings
Field operator	Operations on the ground	Declares harvest/transport/planting activities, records observations, works offline
Company representative	Its own company	Files applications, submits them, pays fees, downloads permits and receipts
Forest explorer	Public data	Browses forests, the map and the Forest Assistant. Read-only.

3.2 What each role sees

The bottom tabs are deliberately different for each role. Hiding a tab is a usability decision only — typing the address of a hidden screen still returns **403 Forbidden**, because the server checks the permission again.

Role	Tabs
Administrator	Home · Permits · Alerts · Map · Profile
Government forest officer	Home · Permits · Field · Alerts · Profile

Role	Tabs
Environmental officer	Home · Map · Field · Alerts · Profile
Forest inspector	Home · Field · Map · Alerts · Profile
Field operator	Home · Field · Map · Sync · Profile
Company representative	Home · Permits · Field · Payments · Profile
Forest explorer	Home · Forests · Assistant · Map · Profile

Everything else lives behind **Home → All modules** (the  button), which is the same for everyone: registers (activities, observations, inspections, violations, reports, companies, users), the audit trail and system settings. Entries you are not entitled to are marked rather than hidden, so you can see what exists.

3.3 Permissions

A permission is a short code, e.g. `permits:approve`, `violations:resolve`, `observations:create`. Two of them shape almost every screen:

- `...:read` — you see **all** records of that kind.
- `...:read_own` — you see only the records linked to **your company** (or created by you).
- `...:read_public` — you see only what is published (forest explorers).

The server decides scope; the app only labels it. **Profile → Roles and scope** shows exactly which scope you have on each module, and **Profile → Permissions** lists the codes you hold. If a button you expect is missing, this is where to look first.

4 The mobile app, screen by screen

4.1 Home

Your start page. It contains:

- **Your identity** — your roles, your company, whether the device is online, and how many records are still queued on this device.
- **Quick actions** — New permit and Capture activity, the two things people do most.
- **Operational picture** (or Forest resources for public accounts) — live counters: forests, permits, active permits, fees invoiced, open alerts, pending payments. Tap any tile to open the matching register, already filtered.
- **Administration** — only if you hold `users:read`, `audit:read` or `settings:manage`: user accounts, companies, the audit trail, system monitoring, system settings.
- **All modules** (, top right) — the full screen index, grouped by theme, with a badge showing how many AI alerts are waiting.

4.2 Map

Forests, zones, protected areas, inventory plots and your own GPS check-ins, drawn from the `/gis` endpoints.

- **Layers** — toggle each feature type on and off. The counter row shows how many features of each type are loaded, and warns when a forest has no boundary polygon (it cannot be drawn until a geometry is imported).

- **Tap a feature** — a sheet with its properties and, where the record exists, an **Open record** button that jumps to the full screen.
- **Search near me** — uses your device position and a radius (1–50 km) to find forests, zones and check-ins around you.
- **Record check-in** — stores your position as a field check-in, with accuracy and whether the fix came from the device GPS.

4.3 Permits

The register of every permit you are allowed to see. Filter by status, type, company, forest or period; sort and paginate.

Open a permit to see its **state machine in practice**:

- Header with the permit number, status pill, type and **DEMO** flag where applicable.
- **Available actions** — the buttons are computed from `GET /permits/{id}/actions`, so you only ever see transitions that are legal from the current status and allowed by your permissions. Some actions ask for a reason (reject, request a revision, suspend, revoke); the reason is stored in the audit trail.
- **Pay the permit fee** — for company representatives on a permit awaiting payment.
- Details: type, purpose, company, forest, zone, validity, volume requested and approved, fee, royalty rate per m³, and the **outstanding balance**.
- **Download the permit** — the permit certificate as a PDF, rendered from the live record (needs `permits:download`).
- **Documents** — attach and verify supporting documents on the permit file.
- **Timeline** — every transition the permit has been through, with who made it and when.

4.4 Payments

Fees, royalties and penalties. **New payment** records a payment and picks up the permit's outstanding balance automatically.

Open a payment for:

- **Provider actions** — Verify with provider re-checks the transaction; in the **simulator** provider you also get Sandbox: mark successful and Sandbox: mark failed so the flow can be demonstrated end to end without moving money.
- **Receipt** — the printable record: issuer, amount, purpose, method, provider reference, payer, company and permit. Sandbox receipts are clearly marked no funds moved.
- The **receipt PDF**, and the history of the payment.

4.5 Field

The working screen for anyone on the ground. Four tiles summarise the day: open inspections, running activities, field observations and, for operators, how many records are queued offline.

- **Capture** — four entry points: Activity, Observation, Inspection, Equipment.
- **Tabs** — Inspections (assigned to you), Activities, Observations. Each row shows its status pill and, for offline records, a **queued / sync** badge and whether the position came from the device GPS.

4.6 Alerts

The AI detection queue (see §7). Each alert carries a severity, an explanation of why it fired, the evidence it was computed from, and the decision controls: **acknowledge**, **dismiss** (with a reason), or **escalate into a violation case** in one tap.

4.7 Assistant

The Forest Assistant: a chat over your own forest data. Ask “which permits expire in the next 30 days?” or “summarise compliance for UFA 09-012” and it answers in French or English, with the sources it used. Conversations are saved (**History**), and each answer shows which engine produced it and how long it took.

4.8 Sync (field operators)

Everything captured while offline lands in a local queue. This screen is the queue: what is pending, what failed and why, and the controls to **Synchronise now**, retry, or discard a record that will never be accepted.

See §6.

4.9 Profile

- **Account** — name, phone, job title (editable in place).
- **Roles and scope** — your roles and, per module, whether you see all, own or public records.
- **Appearance** — light or dark theme.
- **Language** — French or English; affects dates, numbers and status labels everywhere, and the language of generated documents and AI answers.
- **Sessions** — every device currently signed in, with This device marked; revoke any of them individually.
- **Security** — change your password (current password required; changing it signs out every other device).

4.10 Public explore

The **Forest explorer** account — or simply Explore forests on the welcome screen, without signing in at all — gets a read-only view: the forest register, the map, forest detail screens and the Assistant. No permit, payment or compliance data is exposed — every one of those endpoints requires authentication.

5 Step-by-step workflows

Each workflow below is a real path through FEMS. Where an action is only offered to certain roles, the role is named.

5.1 Apply for an exploitation permit — company representative

- 1 **Permits → New permit.**
- 2 Choose the **type** (annual cutting permit, special products, ...), the **forest** and, if the forest is divided, the **zone**.
- 3 Fill in **purpose**, the requested **volume in m³**, and the planned **start and end dates**.
- 4 Save. The permit is created in **DRAFT** — nothing has been sent to the administration yet, and you can still edit it.

- 5 When it is ready, open the permit and press **Submit**. It moves to **SUBMITTED** and becomes read-only for you.
- 6 Watch the status. Officers may ask for a revision (**REVISION_REQUIRED**) with a reason: edit the permit and press **Resubmit**.
- 7 When it reaches **APPROVED**, the fee is calculated and the permit moves to **PAYMENT_PENDING**. Pay it — see 5.3.
- 8 Once the payment is confirmed the officer **activates** the permit. It is now **ACTIVE** and you may start operations against it.

5.2 Review an application — government forest officer

- 1 **Permits**, filter by SUBMITTED.
- 2 Open the application. Check the company's standing, the requested volume against the zone's allowable cut, and the dates.
- 3 Choose an action — the app only shows legal ones:
 - **Start the review** → **UNDER_REVIEW** (takes ownership of the file).
 - **Request a revision** → **REVISION_REQUIRED**, with a reason the applicant sees.
 - **Approve** → **APPROVED**, optionally setting the **approved volume** below what was requested, and confirming the fee and royalty rate.
 - **Reject** → **REJECTED**, with a reason.
- 4 After approval, FEMS raises the fee and moves the permit to **PAYMENT_PENDING**.
- 5 When the payment settles, press **Activate**. The permit becomes **ACTIVE** and its number is issued.

Later, on an active permit you can **Suspend** (with a reason), **Reinstate**, **Revoke** (with a reason), let it **Expire**, or **Renew** it back to a draft.

5.3 Pay the fee and get a receipt — company representative

- 1 **Payments** → **New payment** (or **Pay the permit fee** on the permit itself). The outstanding balance is pre-filled.
- 2 Choose the **method** (mobile money, bank transfer, cash at the treasury, ...) and the **amount**.
- 3 Save. The payment is **PENDING**.
- 4 Open the payment. Verify with provider asks the payment provider for the real status. With the **simulator** provider, use Sandbox: mark successful instead.
- 5 Once **PAID**, the **Receipt** section appears with the receipt number. Open the **receipt PDF** to print or share it.

Unpaid and partially paid permits are visible on the dashboard tile Outstanding, and the permit detail always shows the **outstanding balance** in red while anything is due.

5.4 Declare a harvesting activity — field operator

- 1 **Field** → **Capture** → **Activity** (or Capture activity on Home).
- 2 Choose the **type** (harvest, transport, planting, ...), the **permit** it runs under, the **zone**, and the planned **volume** and **dates**. Add the **equipment** used if relevant.
- 3 Save → **PLANNED**. When the crew actually starts, press **Start** → **IN_PROGRESS**. Starting requires a valid permit; FEMS refuses otherwise.
- 4 If work stops, **Suspend**; when it resumes, **Resume** (checked again against the permit). When the job is done, **Complete** — record the actual volume, species and parcel.

- 5 Attach **photos** on the activity detail screen; they are stored with the record and appear in reports.

5.5 Record a field observation — any field role

The quickest thing to record on site, and the one people do offline.

- 1 **Field → Capture → Observation.**
- 2 Pick a **category** (illegal logging, fire, wildlife, encroachment, erosion, ...), give it a **title** and a description, and set the **severity**.
- 3 Take the **position** — the app offers the device GPS and, if the fix is poor, lets you drop the point on the map instead. Whatever you choose is recorded: **DEVICE_GPS** or **MANUAL**, plus the accuracy in metres. This is what makes an observation usable as evidence later.
- 4 Add **photos** (optional, but strongly recommended).
- 5 Save. If you are offline the observation is queued and badged **QUEUED** until it syncs.

5.6 Run an inspection — forest inspector

- 1 **Field → Capture → Inspection**, or open an inspection assigned to you.
- 2 Set the **target** (permit, zone or activity), the **type** (routine, compliance, incident, verification, post-harvest) and the planned date.
- 3 On site, open the inspection. Press **Get position** and then **Start inspection** — starting is refused without a position, and the fix is stored with its accuracy and the distance from the target. The app warns you if the position looks simulated.
- 4 Work through the **checklist**. The questions depend on the inspection type (routine, compliance, environmental, post-harvest, incident, verification). Answer each one and note what was found. FEMS computes a **compliance score** from the answers.
- 5 Press **Submit inspection**: record the **outcome**, a **summary**, your **recommendations**, the **verified volume** and **verified trees**, and any **discrepancies** between the register and what you found.
- 6 The inspection becomes **SUBMITTED**; a supervisor **reviews** it and then **closes** it. A failed inspection is normally the start of a violation case.

5.7 Open and resolve a violation — environmental officer

- 1 **Violations → New**, or — faster — open an AI alert and press **Escalate to violation**, which carries the evidence and the alert id into the new case.
- 2 Record the **type**, **severity**, the **forest / zone / protected area**, the company and permit concerned, the location, the **estimated damage** and, if you have it, the **evidence** (photos, GPS, observations).
- 3 Work the case with the actions in **Case workflow** — again, only legal ones are offered:
 - **Open an investigation** (**OPEN → UNDER_INVESTIGATION**) — assign it for field verification.
 - **Confirm the violation** (→ **CONFIRMED**) — record the regulatory finding, the assessed damage in XAF and the **penalty**. A reason is required and stored.
 - **Dismiss** (→ **DISMISSED**) — the case was unfounded; say why.
 - **Escalate** (→ **ESCALATED**) — hand it to the hierarchy, prosecution or ministry.
 - **Resolve** (→ **RESOLVED**) — once the penalty is settled and remediation agreed. This needs the dedicated **violations:resolve** permission.
 - **Reopen** — new evidence appeared on a closed case.

- 4 When confirming, set **remediation required** (e.g. replant 120 stems of sapelli in the degraded block) and a **remediation deadline**. FEMS flags the case **remediation overdue** when the date passes.
- 5 The company pays the penalty from the case screen (**Settle the penalty**). The case shows **penalty settled** and can then be resolved.

5.8 Handle an AI alert — officer or administrator

- 1 **Alerts**. New alerts are **NEW**; the badge on **Home → All modules** counts them.
- 2 Open one. Read the **explanation** — the rule that fired, the data it compared and the numbers. An alert you cannot justify is an alert nobody will act on.
- 3 Decide: **Acknowledge** (someone is on it), **Dismiss** with a reason (false positive — this also teaches the engine what to ignore next time), or **Escalate to violation**.
- 4 Alerts you escalated carry an raised from an AI alert badge on the violation case, so the trail is complete in both directions.

5.9 Generate a report

- 1 **Reports**. The catalogue lists built-in reports (permit register, harvest volumes, revenue, violations, inspection coverage, ...) with their parameters.
- 2 Choose one, set the **period** and any **filters** (forest, company, status), and press **Dataset** to preview a few rows before committing.
- 3 Press **Generate** and pick the **format**:
 - **JSON** — hand it to another system or a spreadsheet.
 - **CSV** — open it in Excel.
 - **PDF** — a formatted document, branded with the ministry/organisation from system settings, ready to print or attach to a letter.
- 4 Generated files are listed on the report screen with their parameters, who asked for them and when. Download from there, or **Send** the report to an audience (officers or companies) with a short message — the recipients get a notification with the file attached. Default format is PDF.

6 Working in the field: offline, GPS and photos

6.1 Offline capture

Field operators frequently have no signal in the forest. FEMS is built for that:

- 1 Record your **activities** and **observations** as normal. The app saves them on the device and shows a **Queued** badge instead of pretending they reached the server.
- 2 Home and Field both show how many records are waiting. A full-screen notice appears when the queue is not empty: **Synchronise** sends everything at once.
- 3 Open **Sync** to see the queue in detail: the client reference, the type, the planned volume or category, and the position that was recorded.
- 4 If the server rejects a record (for example, the permit was suspended while you were offline), the record turns red with the server's message, an **attempts** counter, and a **Retry** button. Fix the cause, then retry — or **Discard** the record if it was a mistake.
- 5 **Clear queue** empties the device. It warns you first, because it is the one action that loses data for good.

Rules of thumb:

- Photos and GPS are captured at the moment you press save, so a queued record is complete.
- Sync while you still have some signal — at the edge of the concession is better than at the office two days later.
- A queued record is **not** visible to your colleagues until it syncs. Say so if someone calls asking why the register is behind.

6.2 GPS and evidence quality

Everywhere a position matters (observations, inspection start, check-ins, activity sites), FEMS records three things alongside the coordinates:

- **Accuracy** in metres,
- **Source** — `DEVICE_GPS` , `MANUAL` (you placed the point) or a network fix,
- **Distance from the target**, where there is one.

It also flags a **simulated / mocked** fix, which is how fake GPS apps show up. A record with a mocked position is marked as such, because it will not survive scrutiny as evidence.

Practical advice: wait for the accuracy to settle below ~10 m before saving; if the canopy makes that impossible, say so in the description and place the point manually — an honest manual point beats a wildly wrong automatic one.

7 Artificial intelligence in FEMS

Two features, one principle: **the AI proposes, a human decides.**

7.1 Alerts

A rules engine watches the data you already produce — harvest volumes against permit ceilings, activity outside the authorised zone, missing inspection coverage, repeated observations in the same place, unusual transport volumes — and raises **alerts**.

Every alert carries:

- a **severity** (`LOW` , `MEDIUM` , `HIGH` , `CRITICAL`),
- the **rule** that fired, in plain language,
- the **evidence**: the records and numbers it compared,
- a **state**: `NEW` → `ACKNOWLEDGED` / `DISMISSED` / escalated.

Alerts never change a permit, a payment or a case. Only the three human decisions in 5.8 do.

7.2 The Forest Assistant

A chat over your own forest data, inside **Assistant**. It answers questions like “which permits expire this quarter?”, “what did the last inspection of UFA 09-012 find?” or “summarise violations in the Littoral region this year”, and cites the records it used.

- It only ever reads what **you** are allowed to read — the same scoping rules apply.
- It answers in the language you selected in **Profile → Language**.
- Each reply shows which engine produced it and how long it took. When no external model is configured, a local rule engine answers instead and the badge says so.
- It is an assistant, not an authority: check anything that is going into a decision.

7.3 Analysis

Analysis screens let you ask the engine for a deeper read of a forest, zone or permit — deforestation signals, pressure indicators, compliance posture — and keep the result with its inputs so it can be reviewed later.

8 Reports and documents

Document	Where	Notes
Permit certificate (PDF)	Permit detail → Download the permit	Rendered from the live record, so a re-download after a suspension shows the current status
Payment receipt (PDF)	Payment detail → Receipt	Available once the payment is PAID . Sandbox receipts are marked no funds moved
Report (JSON / CSV / PDF)	Reports	Generated on demand, listed with its parameters and author. See 5.9

Documents are generated server-side from the live data at the moment you ask for them, so a certificate re-downloaded after a suspension shows the current status. Generated files are kept in the backend's `reports-output` directory and listed on the report screen.

9 Administration

Administrator only. All of these are under **Home → Administration** or **All modules**.

9.1 User accounts

- **Users** — search, open, edit, deactivate. Deactivating keeps the history: the account stops working but its audit trail and records stay.
- **Create account** — sets the email, name, role and optional company. The new user gets a verification email; they can be forced to change their password at first sign-in.
- **Unlock and reset** — an account locked by five failed attempts unlocks itself after 15 minutes; to do it now, set its status back to **Active** (that also clears the lock). **Reset password** sets a temporary password and forces a change at the next sign-in. **Restore** brings back an account that was deleted by mistake.

9.2 Roles and permissions

Roles lists the seven built-in roles with their level and member count. Use **Permissions** to see the complete vocabulary the guards use, and to grant or revoke individual permissions on a role.

Be conservative. Permissions are additive and immediate: granting `permits:approve` to a role means every member can approve a permit from the next request onwards.

9.3 System settings

- **Notification delivery** — per notification type, whether it goes to the app (**Push**), by **Email**, or both. In-app delivery is always on and cannot be switched off.
- **Email delivery** — shows the SMTP configuration actually in use (host, port, encryption, account, whether the password is set, the from address) and lets you **send a test email** to any address before relying on it.

- **Other integrations** — payments (provider, currency, webhook) and the AI services, with a **Re-check integrations** button that tests each one and reports the result.
- **Artificial intelligence** — which detector is active, how many rules the engine holds, and whether an external model is configured.

9.4 System monitoring

Settings → System monitoring is the health console. It opens with one verdict — **OK**, **DEGRADED** or **FAILING** — and then the measurements behind it:

- **Process** — uptime, start time, Node version, heap and resident memory.
- **Requests per minute** over the last quarter of an hour, plus the **busiest routes** and the **slowest recent requests**.
- **Scheduled jobs** — the outcome of each one's last run (a failed job shows here before anyone notices missing data).
- **Storage** — driver, directory, number of files and total size.
- **Last 24 hours** — audited actions, successful vs refused sign-ins, notifications created, active sessions.
- **Components** — one line each for the database, mail, payments, AI and the offline sync, green / amber / red, with the reason when it is not green.

Read it at the start of the day. An amber component usually means an integration expired or a scheduled job failed — not that FEMS is broken. The lighter `GET /system/health` is the same verdict without the detail, and the public `/health` probe tells a load balancer only that the process is up.

9.5 The audit trail

All modules → Audit records who did what, to which record, when, from where, with which result. It is append-only and cannot be edited from any screen. When a decision is contested, this is the record that settles it.

10 Using the API directly

Everything above is also a documented HTTP API. This is useful for bulk actions, integrations and troubleshooting.

```
BASE=http://localhost:3000/api/v1

# sign in
TOKEN=$(curl -s -X POST $BASE/auth/login -H "Content-Type: application/json" \
  -d '{"email":"demo.admin@fems.cm","password":"FemsDemo#2026"}' \
  | python3 -c "import sys,json;print(json.load(sys.stdin)['data']['tokens']['accessToken'])")

# use it
curl -s $BASE/permits -H "Authorization: Bearer $TOKEN"
```

Things worth knowing:

- **Base URL** — `http://<host>:3000/api/v1`. Interactive docs at `/docs`, machine-readable schema at `/docs-json`.

- **Tokens** — an access token (15 min) and a refresh token (30 days) come back from `/auth/login` . Send `Authorization:Bearer <access token>` ; refresh with `POST /auth/refresh` .
- **Envelope** — successes are `{ "success":true, "data":..., "meta":{"page, limit, total, totalPages, hasNextPage, hasPreviousPage}, "timestamp":... }` .
- **Errors** — `{ "success":false, "error":{"code":..., "message":..., "details":... } }` . The `code` is the thing to branch on; `details` lists validation failures field by field.
- **Pagination** — `?page=1&limit=20` (max 100), plus `sortBy` / `sortOrder` .
- **Rate limits** — 120 requests/minute, 10/minute on authentication routes. The response carries `X-RateLimit-Limit` , `-Remaining` and `-Reset` .
- **Scope** — a record outside your scope returns **404**, not 403, deliberately: existence is not information a stranger is entitled to.

A ready-to-import **Postman collection** with 194 requests across 17 folders, and 104 screenshots of those requests actually executed, is in `docs/postman/` (`./postman/README.md`) . Import the collection and the environment, run the administrator login, and everything else is pre-filled.

11 Troubleshooting

Symptom	Likely cause	Fix
Invalid credentials on a known-good password	Account locked after 5 failures	Wait 15 min, or have an administrator set the status back to Active
Buttons missing on a permit	Not your permission, or not a legal transition from this status	Check the permit's status and your role scope
<code>403</code> with a permission code in the message	The role lacks that permission	Administrator: grant it, or use an account that has it
<code>404</code> on a record you know exists	It exists but is outside your scope	Sign in with a role that covers it
<code>400 PERMIT_INVALID_TRANSITION</code>	The action is not allowed from the current status	Open the permit and read Available actions
<code>400</code> with a <code>details</code> array	Validation: a field is missing, malformed or out of range	Read <code>details</code> ; it names every offending field
<code>409 ..._CODE_EXISTS</code>	The code (forest, permit, company) is already used	Choose another code, or update the existing record
<code>429</code>	Rate limit	Wait for <code>X-RateLimit-Reset</code> , then retry
Queued records never sync	The server is rejecting them	Open Sync , read the red message on the record
Inspections refuse to start	No GPS fix	Press Get position and wait for accuracy to settle
Position marked simulated	A mock-location app is active	Turn it off; the record will not be usable evidence
No verification / notification email	SMTP not configured or rejected by the provider	Settings → Email delivery → Send test email
Receipt says sandbox receipt — no funds moved	<code>PAYMENT_PROVIDER=simulator</code>	Expected in demonstration environments
The app cannot reach the API on a phone	It is using <code>localhost</code> , or the LAN address changed	Set <code>EXPO_PUBLIC_API_URL</code> to the machine's LAN address
The AI answers oddly or slowly	No external model configured, or the provider is rate-limited	Check Settings → Artificial intelligence ; the badge on each answer names the engine

12 Glossary

Term	Meaning
UFA / Forest management unit	A forest concession identified by a code (e.g. UFA 09-012)
Zone	A management subdivision of a forest: a series, block or parcel
AAC (allowable annual cut)	The volume that may legally be harvested from a zone in a year
Permit	The authorisation to exploit. See the lifecycle below
Activity	A declared operation: harvest, transport, planting, ...
Observation	A field note about something seen on site (illegal logging, fire, ...)
Inspection	A planned visit, scored against a checklist
Violation	A compliance case, from report through penalty to remediation
Royalty	The charge per m ³ harvested, separate from the permit fee
Remediation	Corrective work demanded from an offender (replanting, restoration, ...)
XAF	Central African CFA franc — the currency FEMS records amounts in
isDemo	Flag on seeded demonstration records, shown as a DEMO badge
Scope	Which slice of the data you may see: all, own (your company) or public

Permit lifecycle

Action	From → To	Who	Reason required
SUBMIT	DRAFT → SUBMITTED	applicant	–
START_REVIEW	SUBMITTED → UNDER_REVIEW	officer	–
REQUEST_REVISION	UNDER_REVIEW → REVISION_REQUIRED	officer	✓
RESUBMIT	REVISION_REQUIRED → SUBMITTED	applicant	–
APPROVE	UNDER_REVIEW → APPROVED	officer	–
REJECT	UNDER_REVIEW → REJECTED	officer	✓
MARK_PAYMENT_PENDING	APPROVED → PAYMENT_PENDING	officer / company	–
ACTIVATE	PAYMENT_PENDING → ACTIVE	officer	–
SUSPEND	ACTIVE → SUSPENDED	officer	✓
REINSTATE	SUSPENDED → ACTIVE	officer	–
EXPIRE	ACTIVE / SUSPENDED → EXPIRED	system or officer	–
REVOKE	ACTIVE / SUSPENDED / PAYMENT_PENDING → REVOKED	officer	✓
CANCEL	DRAFT / SUBMITTED / REVISION_REQUIRED → CANCELLED	applicant	–
RENEW	ACTIVE / EXPIRED → DRAFT	applicant	–

Get the same list, filtered to what you may do to this permit, from `GET /permits/{id}/actions` .

Violation lifecycle

`OPEN` → (INVESTIGATE) → `UNDER_INVESTIGATION` → (CONFIRM) → `CONFIRMED` → (RESOLVE) → `RESOLVED` , with `DISMISS` and `ESCALATE` available from the earlier states and `REOPEN` from any closed state. Every transition except `INVESTIGATE` requires a reason, which is stored in the audit trail.

Inspection lifecycle

`SCHEDULED` → (START, needs a GPS fix) → `IN_PROGRESS` → (SUBMIT) → `SUBMITTED` → (REVIEW) → `REVIEWED` → (CLOSE) → `CLOSED` , with `CANCEL` available before submission.

Activity lifecycle

`PLANNED` → (START, requires a valid permit) → `IN_PROGRESS` → (COMPLETE) → `COMPLETED` , with `SUSPEND` / `RESUME` / `CANCEL` available along the way.